
Does Cost-Efficiency Travel?

What Agent Leaderboards Can Predict About Cost Across Workloads

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Abstract

1 Agent leaderboards increasingly report task success and dollar cost, inviting users to choose
2 a “cost-efficient” system from a single benchmark. We ask whether that recommendation
3 travels to a new workload. Using public Holistic Agent Leaderboard (HAL) results, we hold
4 the displayed HAL Generalist Agent scaffold fixed and compare repeated displayed model
5 labels across six workloads. Raw dollar-cost ranks are strongly associated in several high-
6 overlap pairs, but this signal is largely absorbed by stable displayed-label and benchmark
7 cost propensity. Held-out prediction is strong for four workloads and uninformative for
8 two lower-overlap workloads. Aggregate dollars per expected success are less stable: only
9 three CostPerSuccess associations survive permutation testing and Holm correction, and
10 the clearest raw-cost versus CostPerSuccess gap occurs for GAIA-SWE-mini. A separate
11 matched public trace study gives the system-level context that leaderboard aggregates lack:
12 across 977 exact task-model pairs, OpenHands uses about 15% fewer median turns and
13 tool calls than SWE-agent, without a detected difference in resolved status, and the pop-
14 ular “expensive failure” premium dissolves entirely under exact task matching, indicating
15 it measures workload composition rather than failing-run behavior. Taken together, the re-
16 sults show that raw cost is a useful but incomplete portability signal. It combines persistent
17 label-level cost propensity with scaffold execution structure, and it should not be treated as
18 a portable ranking of deployment efficiency.

19 1 The practical problem

20 Agent evaluation is expensive enough that cost has become a first-class leaderboard number. HAL reports
21 21,730 rollouts across nine benchmarks and roughly \$40,000 in evaluation cost [Kapoor et al., 2026]. That
22 makes a cost-accuracy frontier appealing: if two systems perform similarly, why not choose the cheaper one?

23 The hidden step in that reasoning is portability. A user sees that a label is inexpensive on one benchmark
24 and assumes it will remain a good buy in a different environment. That assumption is plausible, but it is not
25 innocuous. A reported dollar cost can reflect the model or provider, reasoning mode, prompt and tool setup,
26 cache behavior, scaffold, and the workload itself.

27 The GAIA-SWE-mini comparison makes the issue concrete. Across 17 shared displayed labels, raw dollar-cost
28 ranks are closely aligned (Spearman $\rho = 0.89$). But the ranking by aggregate dollars per expected success is
29 much less aligned ($\rho = 0.29$); the paired raw-minus-CPS contrast is 0.60 with a 95% configuration-resampling
30 sensitivity interval of [0.14, 1.13]. In practical terms, a GAIA-like leaderboard can offer a useful prior about
31 raw spend for a SWE-like workload without providing a reliable ordering of expected dollars per successful
32 task. This is an aggregate ranking example, not a claim about a named model or a production deployment.

33 We therefore ask three questions. First, when does an observed cost ranking predict a held-out workload? Sec-
34 ond, does that ranking still travel after separating stable displayed-label cost propensity from benchmark cost
35 level? Third, what can public traces say about the scaffold-level execution structure that aggregate leaderboards
36 hide?

2 Why this is different from prior cost and benchmark work

This paper is an empirical evaluation study, not a new rank estimator. Kapoor et al. [2024] established the importance of reporting accuracy and cost jointly and motivates the frontier perspective used here. Ndzomga [2026] asks whether a smaller task subset preserves agent rankings within a benchmark; we instead ask whether an observed cost signal predicts a different workload. Benchmark² studies cross-benchmark rank consistency for static LLM benchmarks; our setting adds interactive scaffolds, dollar cost, success adjustment, and a held-out workload test. SWE-Effi examines resource effectiveness within software engineering and reports expensive failures; our matched design tests that heuristic at fixed task and finds no failure premium. The matched Open-SWE extension does not claim to reproduce HAL cost measurements. It supplies separate, exact task–model evidence that scaffold choice can change observable trajectory burden.

Our primary data are the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026]: 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 displayed model labels. The coverage determines the design. Most scaffolds appear on one workload only, so the data do not support a general model–scaffold–workload decomposition. We instead fix the displayed HAL Generalist Agent scaffold, match exact public model-label strings, and retain six workloads with enough repeated labels: CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, and TAU-bench Airline. This cohort contains 86 rows. USACO has one Generalist observation and is excluded.

The primary CSV supplies accuracy and aggregate dollar cost, but not token components, provider aliases, API calls, tool calls, retries, cache usage, or per-task trajectories. Accordingly, we do not call any adjustment “price controlled,” and we do not infer token or failure costs from dollars. Most evaluations are single-run, so rollout-level uncertainty is unavailable.

For each pair with at least five shared labels, we rank the same shared displayed labels by each metric. A configuration-resampling sensitivity interval is obtained by drawing N shared labels with replacement, keeping each draw’s paired values together, and recomputing the statistic; 10,000 draws yield its 2.5th and 97.5th percentiles. It describes sensitivity to the finite observed label set, not a rollout-level confidence interval. Raw-cost Holm correction applies to 15 tests. CPS has its own 20,000-draw label-pairing permutation tests and Holm correction over 15 tests.

$$\text{CPS}_{i,b}(\epsilon) = \frac{C_{i,b}}{\max(A_{i,b}, \epsilon)},$$

where A is accuracy as a fraction and the primary floor is $\epsilon = 0.01$. The 5% and zero-exclusion versions are reported in the supplement. CPS is an aggregate dollars-per-expected-success proxy, not the mean cost of successful runs.

To test out-of-sample portability, we use leave-one-benchmark-out (LOBO) prediction. For each held-out workload, we estimate each label’s centered log-cost propensity from the other workloads and predict the held-out cost rank. We also fit the descriptive pooled model

$$\log C_{i,b} = \mu + \alpha_i + \beta_b + \epsilon_{i,b}.$$

This 86-row model has 30 parameters and 56 residual degrees of freedom. Its fit is descriptive; LOBO is the stronger test of portability.

Finally, we report three decision frontiers: discrete nondominance, a HAL-inspired origin-anchored convex hull, and a 5% near-tie tolerance. These answer related but different decision questions.

3 Evidence

3.1 Raw cost predicts some held-out workloads

The most direct evidence comes from LOBO. A label’s cost propensity learned from the other workloads predicts held-out raw cost ranks for CORE ($\rho = 0.78$, $N = 14$, $p = 0.0016$), GAIA (0.87 , $N = 16$, $p < 0.0001$), SWE-mini (0.90 , $N = 16$, $p < 0.0001$), and TAU (0.83 , $N = 14$, $p = 0.0004$). These are not one-label artifacts: leave-one-label-out ranges remain positive for every omission, from $[0.73, 0.91]$ for CORE to $[0.88, 0.92]$ for SWE-mini.

The picture is not uniform. SciCode is unstable under leave-one-label-out analysis ($[-0.50, 0.14]$). ScienceAgentBench remains directionally negative ($[-0.77, -0.43]$), but it has seven labels and its original permutation result is not significant. SciCode has very low mean accuracy (2.1%) and one-third zero-accuracy labels; ScienceAgentBench has the narrowest log-cost dispersion. These are useful diagnostic facts, not a basis for a domain-level causal story.

86 3.2 What the raw-cost signal appears to contain

87 Among high-overlap pairs, raw dollar-cost correlations range from 0.74 to 0.93. A pooled benchmark-plus-label log-cost model has $R^2 = 82.0\%$ and adjusted $R^2 = 72.7\%$, but this in-sample fit should not be overread
88 given 30 parameters and 86 rows. Its purpose is diagnostic. Once observed benchmark and displayed-label cost
89 propensity are removed, none of the high-overlap residual rank associations survives a 20,000-draw permutation
90 test.
91

92 The reasonable interpretation is not that cost is meaningless. Rather, much of the portable raw-cost signal
93 is a stable observed label-level component. The public data cannot identify whether that component comes
94 from provider pricing, reasoning settings, routing, token behavior, cache treatment, or other unrecorded setup
95 choices.

96 3.3 Cheapness is not the same ranking as dollars per expected success

97 CPS changes the question from “which label is cheap?” to “which label buys expected success cheaply?” Under
98 the 1% floor, CPS association survives 20,000-draw permutation testing and Holm correction for CORE–GAIA
99 ($\rho = 0.87$, 95% sensitivity interval [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), and GAIA–TAU (0.78, [0.42,
100 0.94]). Most other CPS associations remain uncertain.

101 The high-overlap mean CPS correlation is 0.59, compared with 0.84 for raw cost, but that average is descriptive
102 rather than a population-level contrast. The clearest paired difference is GAIA–SWE-mini: raw cost exceeds
103 CPS correlation by 0.60, with a 95% resampling interval [0.14, 1.13]. In short, a raw-spend ranking can travel
104 while a success-adjusted ranking does not.

105 3.4 There is no universal cost–accuracy winner

106 No broadly tested label lies on every frontier under nondominance, the HAL-inspired hull, or the 5% tolerance
107 rule. This is a practical guardrail, not the paper’s central novelty. The stronger statement that no label appears
108 on more than 4/6 frontiers fails under tolerance: o4-mini Low appears on 5/6.

109 4 A separate matched trace study

110 The leaderboard analysis tells us what travels, but not what happens inside an agent run. HAL public trace
111 archives are encrypted. We therefore use a separate public Open-SWE-Traces boundary sample: four docu-
112 mented MiniMax-M2.5 shards under SWE-agent and OpenHands. It is not a random sample of the full corpus.
113 We exactly match task ID and model, exclude duplicate task/model/scaffold records, and obtain 977 pairs.

114 The traces expose resolved status, structured messages, tool calls, and text. They do not expose billing, tokens,
115 provider usage, retries, or latency. We therefore call turns, tool calls, and message characters trace-burden
116 proxies, not cost.

117 At fixed task and model, OpenHands/SWE-agent has a median ratio of 0.85 for trajectory turns (10,000-
118 bootstrap interval [0.82, 0.87]) and 0.85 for tool calls ([0.82, 0.86]). OpenHands has lower values on 69.7% of
119 pairs for each measure. Total trajectory characters have a smaller ratio effect, 0.95 ([0.92, 0.98]), with Open-
120 Hands lower on 55.1% of pairs. Resolved status has no detected paired difference ($p = 0.197$). Thus, scaffolds
121 can change visible execution burden even when task and model are held fixed. This does not show lower token
122 use or dollar cost, and it does not identify the mechanism behind HAL rankings.

123 The same matched design tests the popular “expensive failure” heuristic. Across tasks, failed trajectories look
124 19–36% heavier than resolved ones, matching prior resource-effectiveness claims. But among the 107 pairs
125 where the two scaffolds disagree on the identical task, the failed side is heavier only 47–52 times out of 107
126 on every metric ($p \geq 0.25$), and after normalizing by each scaffold’s median burden, failed/resolved median
127 ratios are 0.98–1.06 with intervals spanning 1 (Figure 1). Hard tasks make agents both burn more and fail;
128 unconditioned failure-cost ratios measure workload composition, not failing-run behavior, and inherit exactly
129 the composition problem documented above for raw aggregate dollars.

130 5 Interpretation and implications

131 The evidence supports a narrower conclusion than a single leaderboard recommendation suggests. Raw dollar
132 cost can be predictive across some workloads. In the HAL cohort, that prediction is robust for four held-out
133 workloads and is not driven by a single displayed label. But the signal is partly stable label-level propensity, not
134 a demonstrated measure of portable execution efficiency. Success adjustment changes the ranking and is only
135 partly portable. The matched trace study supplies a compatible systems-level explanation: scaffold choices alter
136 observable work even when task and model are fixed, and the apparent failure-cost premium disappears under

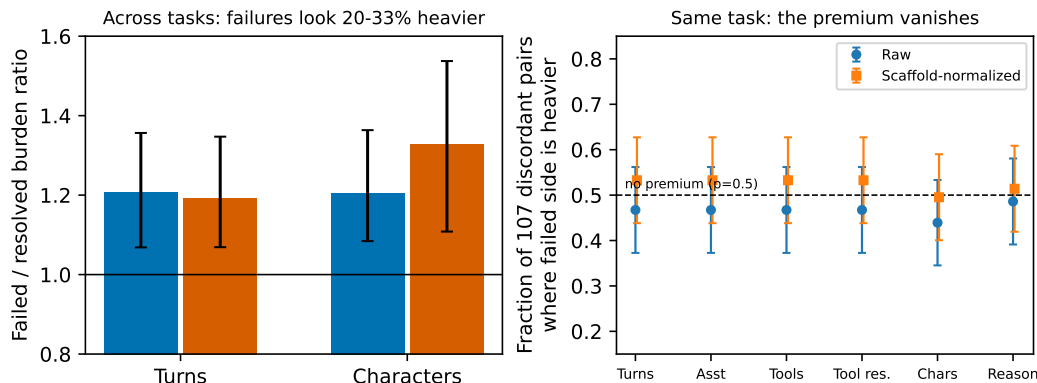


Figure 1: Failure burden is a composition artifact. Left: across tasks, failed trajectories appear 20–33% heavier. Right: on 107 matched pairs where scaffolds disagree, the failed side is heavier about half the time, raw or normalized, consistent with no premium.

137 exact task matching. One composition problem runs through every result: unconditioned aggregates — raw
 138 cost ranks, success-adjusted efficiency, failure premiums — mostly measure which workloads were attempted,
 139 not how agents behaved on them.

140 For practitioners, this means that a leaderboard’s dollar figure is useful as a first screen, not as a final procure-
 141 ment decision. A target evaluation should include a success-adjusted metric and, where possible, trajectory
 142 telemetry. For benchmark builders, the missing fields are concrete: model/provider aliases, input/output/cache
 143 usage, calls, tool activity, retries, outcome labels, and per-task costs would make it possible to distinguish price
 144 from execution behavior.

145 6 Limits

146 This is a secondary-data analysis. Public display labels do not fully identify prompts, tools, budgets, harness
 147 versions, or routing. The HAL cohort is sparse and unbalanced, with mostly single-run rows. The public
 148 HAL snapshot cannot identify failure-cost ratios, retry behavior, token trajectories, tail cost, or a causal model-
 149 scaffold-workload decomposition. CPS is not task-level successful-run cost. The Open-SWE extension is
 150 external to HAL and uses a reproducible four-shard boundary sample, not the full corpus. Its proxies are not
 151 tokens or dollars. These limits define the claims rather than invalidate them.

152 7 Conclusion

153 A cost-aware leaderboard is better than an accuracy-only leaderboard, but it is not a portable deployment-
 154 efficiency ranking by default. In these public agent data, raw cost can generalize to some held-out workloads,
 155 yet it partly reflects stable displayed-label propensity and does not always preserve the ranking by dollars per
 156 expected success. Matched traces show that scaffold design changes observable execution burden at fixed
 157 task and model, while the popular failure-cost premium dissolves entirely under exact task matching. The
 158 unconditioned aggregates that leaders and procurement rules naturally compute — raw cost ranks, success-
 159 adjusted ratios, failure premiums — all measure workload composition more than agent behavior. The next
 160 generation of agent leaderboards should make these distinctions measurable rather than leaving them hidden
 161 inside one dollar number.

162 References

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